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Original article

An ensemble machine learning-based approach to predict cervical cancer using hybrid feature selection

Khandaker Mohammad Mohi Uddin [a,*], Abdullah Al Mamun [b], Anamika Chakrabarti [b], Rafid Mostafiz [c], Samrat Kumar Dey [d]

[a] Department of Computer Science and Engineering, Southeast University, Bangladesh
[b] Department of Computer Science and Engineering, Dhaka International University, Bangladesh
[c] Institute of Information Technology, Noakhali Science and Technology University, Bangladesh
[d] School of Science and Technology, Bangladesh Open University, Bangladesh

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ABSTRACT

Cervical cancer has recently emerged as the leading cause of premature death among women. Around 85% of cervical cancer cases occur in underdeveloped countries. There are several risk factors associated with cervical cancer. This study describes a novel predictive model that uses early screening and risk trends from individual health records to forecast cervical cancer patients' prognoses. This study uses machine learning classification techniques to investigate the risk factors for cervical cancer. Additionally, use the voting method to evaluate all models and select the most appropriate model. The dataset used in this study contains missing values and shows a significant imbalance. Thus, the Random Oversampling technique was used as a sampling method. We used Principal Component Analysis (PCA) and XGBoost feature selection techniques to determine the most important features. To predict the accuracy, we used several machine learning classifiers, including Support Vector Machines (SVM), Random Forest (RF), k-nearest Neighbors (KNN), Decision Trees (DT), Naive Bayes (NB), Logistic Regression (LR), AdaBoost (AdB), Gradient Boosting (GB), Multilayer Perceptron (MLP), and Nearest Centroid Classifier (NCC). To demonstrate the efficacy of the suggested model, a comparison of its accuracy, sensitivity, and specificity was performed. We used the Random Oversampling approach along with the Ensemble ML method, hard voting on RF and MLP, and achieved 99.19% accuracy. It is demonstrated that the ensemble ML classifier (hard voting) performs better at handling classification problems when features are decreased and the high-class imbalance problem is handled.

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1. Introduction

According to the World Health Organization (WHO), women's cancers such as ovarian, bone, breast, and cervical cancer are the leading causes of unseasonable female death [1]. Cervical cancer, also known as cervix cancer, is a malignant tumor that develops when the tissue cells wrapping around the cervix grow and multiply uncontrollably without following the normal cell division process [2]. Cervical cancer is expected to be the third most common disease among women worldwide by 2020, with 6,04,000 new cases reported. Nearly 90% of the expected 34,200 cervical cancer deaths in 2020 will occur in middle- to low-income countries, and the risk of cervical cancer is six times higher for women with HIV than for non-HIV-positive women, with HIV accounting for 5% of all cervical cancer cases [3,4]. Furthermore, HIV causes an abnormally high rate of cervical cancer worldwide. The primary risk factor for cervical cancer is a lifetime infection with positive high-risk types of HPV. Human Papillomavirus (HPV) is a group of viruses that can infect the skin and mucous membranes. There are over 200 different types of HPV, which are divided into low-risk and high-risk categories. Low-risk HPV can cause common skin or genital warts, but they are rarely associated with cancer. High-risk HPV types have been linked to a variety of cancers, including cervical cancer and cancers of the vulva, vagina, penis, anus, and throat. HPV is most commonly transmitted through intimate skin-to-skin contact, but it can also spread through other forms of close contact. Vaccines are available to prevent infection with the most common high-risk types of HPV, significantly lowering the risk of associated cancers. That is thought to be the intention behind al-

* Corresponding author.
E-mail addresses: jilanicsejnu@gmail.com (K.M.M. Uddin), almamundiu16@students.diu.ac (A. Al Mamun), anamikaonuc@students.diu.ac (A. Chakrabarti), rafid.iit@nstu.edu.bd (R. Mostafiz), samrat.sst@bou.ac.bd (S.K. Dey).

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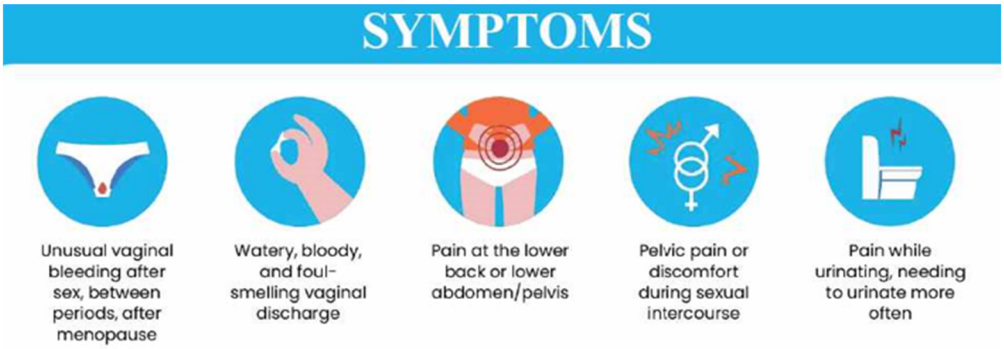

Fig. 1. Cervical Cancer: Symptoms.

most all cases of cervical cancer. Smoking is another major risk factor for cervical cancer. There is compelling evidence to support the claim that women who have used the tablet for five years or more are significantly more likely to develop cervical cancer in HPV-positive individuals. Similarly, taking the pill appears to reduce the risk of uterine and ovarian cancers. Smoking and passive smoking have other negative effects, such as a weakened immune system. At some point in their lives, approximately 8 out of 10 women with an HPV infection are at risk of developing cervical cancer; however, most HPV infections do not lead to cervical cancer [5,6]. In 2023, 13,960 US women may be diagnosed with invasive cervical cancer, with a 50% decrease in prevalence costs since the mid-1970s. The death rate is expected to rise, with 4,310 deaths in the US. The average age of diagnosis is 50, with more than 20% discovered after the age of 65 [7]. A wealth of knowledge about cancer has been accumulated as a result of advancements in medical technology, and it is now readily available to a wide range of medical researchers [8]. Machine learning researchers are constantly striving to improve the cervical cancer data that is now available and can be studied using predictive models. Predictive models developed using machine learning have accelerated the diagnosis of cervical cancer [9]. However, one or more of the following challenges continue to limit these models' ability to predict cervical cancer outcomes: The use of dimension reduction techniques, resampling and data balancing methods to address skewed data, and the resolution of overfitting in Decision Tree (DT) problems are all absent. Dimensionally Reduced Methodologies is a term refers to techniques used to reduce the number of features or variables in a dataset while retaining its essential characteristics. Common methods include Principal Component Analysis (PCA) and feature selection techniques. Decision tree is a type of model used in data analysis and machine learning [10–12]. Furthermore, an innovative approach to developing a machine learning model may provide the opportunity to combat cervical cancer more effectively and improve the future for girls and women. Based on the patient's risk variables and preliminary screening findings from each person's medical data, this work proposes a machine learning model for projecting how cervical cancer may progress in a patient test. This work makes a significant contribution to the field of healthcare by presenting a prediction model that can aid in the accurate identification of cervical cancer using ML approaches. Other cancers, such as blood, breast, and postal cancers, can also be diagnosed using the described technique in a medical setting. The primary goal of this study is to forecast the outcomes of biopsies performed on cervical cancer patients. When evaluating the model's performance, sensitivity takes precedence over other metrics such as accuracy because it allows for a more positive cervical cancer test result for patients who are already incontinent.

Fig. 1 shows Cervical Cancer: Symptoms.

The following noteworthy contributions have been contributed to the current study. A straightforward, accurate, and efficient machine-learning-based cervical cancer prediction model has been developed.

- To acquire a deeper understanding of their role in the risk prediction of cervical cancer, different feature representations were applied.
- Missing value management, data encoding, feature selection, and resampling are the processes in data preprocessing.
- The ROS technique is used to balance data and avoid bias.
- Optimal feature selection is done by utilizing XGBoost, SelectKbest, PCA, or a combination of XGBoost and PCA.
- By deploying numerous classifiers instead of a single machine learning method, ensemble learning guarantees the correctness of the prediction model.

2. Related work

This section focuses on different prediction algorithms presented in recent years for predicting cervical cancer, categorizing them as machine learning-based approaches.

Jiayi Lu et al. [13] introduced an innovative ensemble technique tailored to estimate the risk of cervical cancer. Their approach addresses shortcomings in traditional voting-based historical analyses by advocating for a data rectification process to enhance prediction accuracy. Additionally, they explore the potential benefits of gene assistance modules in refining predictability. Through rigorous measurements, their method, utilizing the voting approach, demonstrates promising accuracy in predicting cervical cancer risk.

Nithya et al. [14] aimed to deepen understanding of cervical cancer risk factors by leveraging machine learning (ML) techniques in the R environment. They systematically investigated various feature selection techniques to identify pivotal elements for accurate prediction. Their approach involves iterative cycles of model training, employing diverse feature selection methods to uncover essential features and establish an optimal feature selection model.

Akter et al. [15] employed three ML models—Decision Trees (DT), Random Forests (RF), and XGBoost—to forecast cervical cancer risk based on behavioral and feature data. Their study showcases significant performance improvements over existing techniques, achieving an impressive 93.33% accuracy.

Asadi et al. [16] reaffirmed the potential of ML in enhancing cervical cancer prediction. Their findings underscore the effectiveness of Decision Tree (DT) algorithms in discerning crucial predictors, contributing to improved predictive accuracy.

Sujay et al. [17] applied ML methodologies to expedite and refine the determination process, particularly in swiftly identifying cancerous samples. By leveraging biopsy data with high resolution,

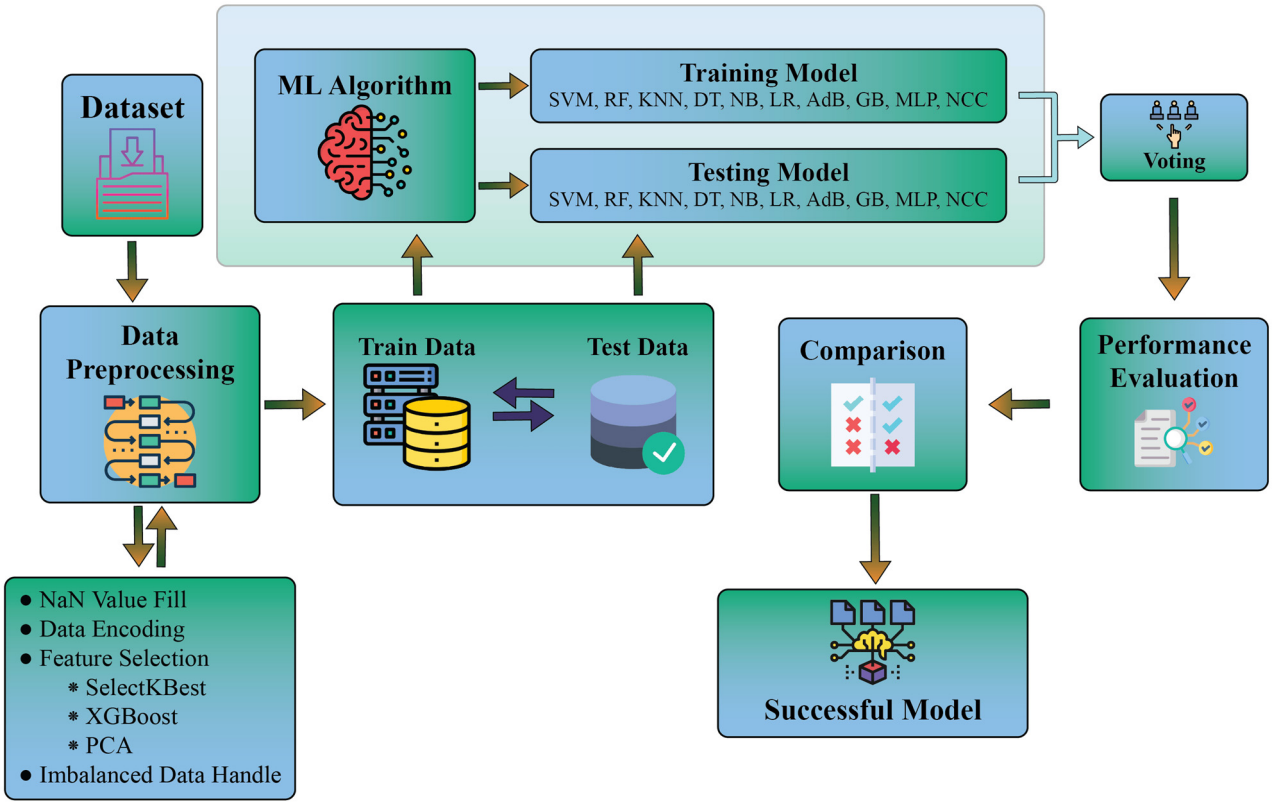

Fig. 2. The visual representation of the proposed method.

their study showcases the Bayes Net approach, achieving an impressive Area Under the Curve (AUC) of 95% and an accuracy of 96.38% after training various ML models.

I. J. Ratul et al. [18] presented supervised ML techniques aimed at anticipating early signs of cervical cancer using a dataset from the UCI Machine Learning repository. Their study rigorously evaluates performance measures including Accuracy, Recall, F1-score, Precision, and ROC-AUC, providing valuable insights into early threat prediction.

Tanimu et al. [9] proposed a decision tree classification approach for analyzing cervical cancer risk factors. Through meticulous examination of Recursive Feature Elimination (RFE), Least Absolute Shrinkage and Selection Operator (LASSO), and other feature selection methods, they identify pertinent qualities crucial for cervical cancer identification.

3. Materials and method

Fig. 2 shows the complete workflow diagram for the proposed paradigm. At first, data from the dataset was pre-processed for missing value handling and data encoding to convert categorical variables into numerical values. Several feature selection approaches are used to extract the most important features from the dataset. A resampling method is used to optimize the dataset. Following that, we divided the dataset into 80% for training and 20% for testing, respectively. Following that, various machine-learning techniques are used on the training data to train the models. After that, run the models through the test dataset before applying the voting classifier. The confusion matrix is used to assess the effectiveness of our model. Finally, we compare the results to determine the best model.

3.1. Dataset description

The Hospital 'Universitario de Caracas' in Caracas, Venezuela, donated the information for this dataset. The Risk Factors dataset may be found on the UCI ML repository [19] and it's a public dataset with 858 patient samples in 36 attributes. There are 36 features in this dataset but the main features are Age, Number_of_sexual_partners, First_sexual_intercourse, Num_of_pregnancies, Smokes, Smokes_years, Smokes_packs_per_year, IUD, IUD_years,..... Dx_Cancer, Dx_CIN, Dx_HPV, Dx, Hinselmann, Schiller, Citology, Biopsy. But Hinselmann, Schiller, cytology, and biopsy are our main target features. In this study, we use Biopsy as our main target outcome. With the use of a biopsy, Fig. 3 shows the class distribution of patients, where 803 are non-cancerous and 55 are cancerous.

3.2. Dataset preprocessing

To achieve maximum accuracy these datasets, need to be processed before being fitted into ML models. Pre-processing techniques are used to address problems such as missing numbers, outliers, label encoding, and others. Missing values can be found in this dataset and before applying the Machine Learning model it needs to be cleaned. The mean and median of the column's non-missing values are used as imputations to fill in the blanks when a column contains missing values. The number of dimensions in the training dataset is reduced using PCA. The most crucial variables in the dataset are chosen using the feature selection method. Fig. 5 represents the numerical column representation of Age, Number_of_sexual_partners, Num_of_pregnancies, Smokes, Dx_Cancer, Dx, Hinselmann, Schiller, Citology, those are the most important values from the dataset.

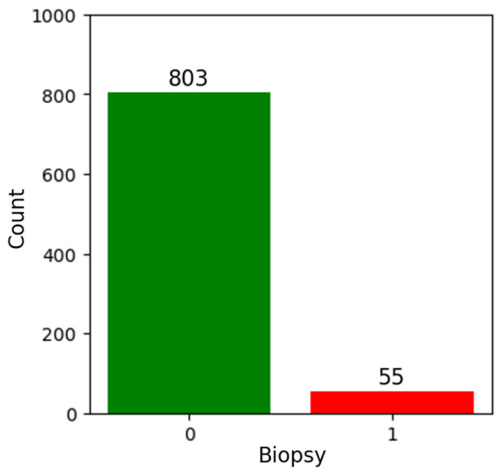

Fig. 3. Cancerous and non-cancerous patients.

Table 1
Important Attributes Selected by PCA.

Priority	Features
1	*Citology*
2	*Dx*
3	*Dx_Cancer*
4	*Dx_CIN*
5	*Dx_HPV*
6	*Hinselmann*
7	*Schiller*
8	*STDs_AIDS*
9	*STDs_cervical_condylomatosis*
10	*STDs_genital_herpes*
11	*STDs_Hepatitis_B*
12	*STDs_HPV*
13	*STDs_molluscum_contagiosum*
14	*STDs_Number_of_diagnosis*
15	*STDs_pelvic_inflammatory_disease*
16	*STDs_syphilis*
17	*STDs_Time_since_first_diagnosis*
18	*STDs_Time_since_last_diagnosis*
19	*STDs_vaginal_condylomatosis*
20	*Biopsy*

3.2.1. Feature selection

To increase accuracy, feature selection is employed in the ML process. Selecting the most crucial variables and eliminating the redundant and unnecessary ones, also increases the algorithm's ability for prediction. In this study, we use PCA, SelectKBest, and XGBoost to pick the most important variables.

a) PCA

Principal component analysis, a popular technique, is used to analyze large datasets with a high number of characteristics per observation to improve the interpretability of the data while preserving the most information. The ability to view multidimensional data is another PCA benefit. PCA is a formal statistical technique for reducing a dataset's dimensionality [20]. Table 1 shows the attributes selected through the PCA feature selection approach.

b) XGBoost

In the machine learning process, feature selection is utilized to increase accuracy. The algorithm's ability to forecast is further enhanced by selecting the most crucial variables and eliminating the redundant and unnecessary ones. We utilized XGBoost to determine which factors were most important for our investigation. XGBoost is a potent machine-learning algorithm based on gradient-boosting decision trees. The basic idea is to combine the predicted results from many separate decision tree models that have undergone multiple training rounds. The approach prioritizes samples with high error rates. It alters the contribution of each underlying decision tree model by weighting it to gradually improve performance at each step of the model [21]. The features chosen using the XGBoost feature selection approach are shown in Table 2.

Table 2
Selected Features (XGBoost).

Priority	Attributes	Importance
1	*Schiller*	*0.372705*
2	*Hinselmann*	*0.160002*
3	*Citology*	*0.076914*
4	*Dx_Cancer*	*0.055241*
5	*STDs_syphilis*	*0.052083*
6	*Dx*	*0.036602*
7	*Dx_HPV*	*0.029568*
8	*STDs_Time_since_first_diagnosis*	*0.026331*
9	*Num_of_pregnancies*	*0.025440*
10	*Smokes_packs_per_year*	*0.022946*
11	*Hormonal_Contraceptives_years*	*0.018842*
12	*STDs_Number_of_diagnosis*	*0.017675*
13	*STDs_number*	*0.016101*
14	*IUD_years*	*0.013764*
15	*STDs_Time_since_last_diagnosis*	*0.012161*
16	*Age*	*0.010484*
17	*Smokes_years*	*0.010361*
18	*Number_of_sexual_partners*	*0.008126*
19	*Hormonal_Contraceptives*	*0.007358*
20	*STDs*	*0.006977*

Table 3
Important Features Using SelectKBest.

Priority	Features
1	*STDs_cervical_condylomatosis*
2	*STDs_vaginal_condylomatosis*
3	*STDs_syphilis*
4	*STDs_pelvic_inflammatory_disease*
5	*STDs_genital_herpes*
6	*STDs_molluscum_contagiosum*
7	*STDs_AIDS*
8	*STDs_Hepatitis_B*
9	*STDs_HPV*
10	*STDs_Number_of_diagnosis*
11	*STDs_Time_since_first_diagnosis*
12	*STDs_Time_since_last_diagnosis*
13	*Dx_Cancer*
14	*Dx_CIN*
15	*Dx_HPV*
16	*Dx*
17	*Hinselmann*
18	*Schiller*
19	*Citology*
20	*STDs_cervical_condylomatosis*

c) SelectKBest

SelectKBest is a feature selection method that chooses features based on the highest scores of a given scoring function. As a scoring function, the sci-kit-learn library's mutual_info_classif is employed. It ranks characteristics based on their interdependence [22]. The important features chosen using the SelectKBest feature selection approach are shown in Table 3.

c) XGBoost + PCA

Combining XGBoost with PCA can be a strong tool for improving the performance of our machine-learning model. PCA is commonly used for dimensionality reduction, which can assist in alleviating the curse of dimensionality and improve XGBoost's efficiency. Use PCA to minimize the number of dimensions in the feature space. This entails computing the main components and picking a subset of them based on the desired level of variance retention. After that, train an XGBoost classifier with the PCA-transformed features. Table 4 shows the attributes selected

Table 4
Important Features Using XGBoost + PCA.

Priority	Features
1	Dx
2	Schiller
3	Citology
4	Hinselmann
5	STDs_AIDS
6	Dx_CIN
7	Dx_HPV
8	STDs_HPV
9	Smokes_packs_per_year
10	Smokes_years
11	Dx_Cancer
12	STDs_vaginal_condylomatosis
13	Hormonal_Contraceptives
14	STDs_molluscum_contagiosum
15	STDs_cervical_condylomatosis
16	IUD_years
17	STDs_Number_of_diagnosis
18	STDs_Time_since_first_diagnosis
19	STDs_Time_since_last_diagnosis
20	*Biopsy*

through the XGBoost + PCA feature selection approach. Fig. 4 shows the correlation between the selected features.

3.2.2. Imbalanced data handling

This dataset has 803 patients who are non-cancerous and 55 patients are cancerous. An oversampling process is used to handle this dataset and Fig. 6 shows the visualization of balanced data. This method's main goal is to replicate minority class samples at random [23]. Replicating all of the initial samples, the ROS increases the dataset. The essential component is that sample variety is maintained since the ROS doesn't produce fresh samples [24].

The number of patients with cancer was matched to the number of patients without cancer, resulting in 803 patients in each group. It is crucial to remember that this modification was made for analytical reasons. Data is frequently unbalanced in real-world situations, and balancing techniques are used to correct this imbalance.

In order to balance the dataset, duplicate samples from the original data are created via random oversampling. Although this method can help address class imbalances, because the oversampled data lacks variability, it may cause overfitting, a situation in which the model performs well on training data but poorly on unseen data. It is crucial to take into account additional strategies like data augmentation, sophisticated oversampling techniques (like SMOTE), or ensemble methods that can lessen overfitting and enhance model generalization in order to address this problem.

Since random oversampling was used to balance the cancerous data with the non-cancerous dataset, there is a risk of data leakage, where the same data points may inadvertently appear in both the training and testing subsets. To address this, it is crucial to ensure that the training and testing data are completely separated.

3.3. Machine learning classifier

In this study, performance has been evaluated using 10 classifiers which are SVM, RF, KNN, DT, NB, LR, AB, GB, NCC, and MLP. Following that, Voting is used for the best model having the best accuracy of all classifiers.

3.3.1. Support vector machine

A support Vector Machine is a supervised learning algorithm that is used for classification and regression applications. Its major goal is to select the appropriate hyperplane for separating data points into distinct classes [25]. Through the use of kernel functions, SVMs work well in high-dimensional spaces and are successful for both linearly and non-linearly separable datasets [26].

The fundamental concept underlying the support vector machine is to identify a hyperplane that maximizes the margin—that is, the distance between the hyperplane and the closest data points from each class, or support vector [27].

Equation (1) represents the linear SVM classifier.

$$f(a) = sign(x \cdot a + b) \tag{1}$$

Here,

- $f(a)$ is the decision function that predicts the class of the input a.
- x is the weight vector perpendicular to the hyperplane.
- a is the input feature vector.
- b is the bias or intercept term.

The hyperplane or decision boundary dividing the classes is represented by the equation $\{(x \cdot a) + b\} = 0$ [28]. Data points nearest to this hyperplane are called support vectors.

3.3.2. Random forest

RF is a group of decision trees that use the standard of several samples to increase the level of accuracy of a provided dataset. It can tackle challenges with classification and regression [29]. In comparison to a single decision tree, the Random Forest approach decreases overfitting and improves accuracy by utilizing the experience of users to merge several weak models into a single, stronger model [30]. Equation (2) represents the RF classifier.

$$R(a) = \sum_{R=1}^{A} X_R \Pi(\mathrm{i} \in B_R) = \sum_{R=1}^{A} X_R \emptyset(i; z_R) \tag{2}$$

Basic decision tree in RF where X_R is the Rth region's typical reaction. B_R, Z_R is the parameter that is divided on, and $\phi(i; Z_R)$ is the limit of information.

3.3.3. K-Nearest Neighbor

Among supervised learning techniques, the non-parametric KNN method detects similarities in the training set [31]. For classification and regression applications, the KNN technique is a popular and straightforward supervised learning algorithm. It is an instance-based, non-parametric algorithm that categorizes new instances according to how similar they are to the ones that already exist in the training dataset [32]. Equations to 5 represent the KNN classifier.

$$\text{Euclidean} = \sqrt{\sum_{i=1}^{a} (b_i - c_i)^2} \tag{3}$$

$$\text{Manhattan} = \sum_{i=1}^{a} \mid b_i - c_i \mid \tag{4}$$

$$\text{Minkowski} = \left(\sum_{i=1}^{a} (\mid b_i - c_i \mid)^d \right)^{\frac{1}{d}} \tag{5}$$

KNN is commonly based on some distance algorithms and those are Euclidian, Manhattan, and Minkowski. Those algorithms evaluate the separation between the present data and all other data. Where b and c are two instances in the dataset, a is the number of features, and b_i and c_i are the values of the ith feature in instances b and c, respectively.

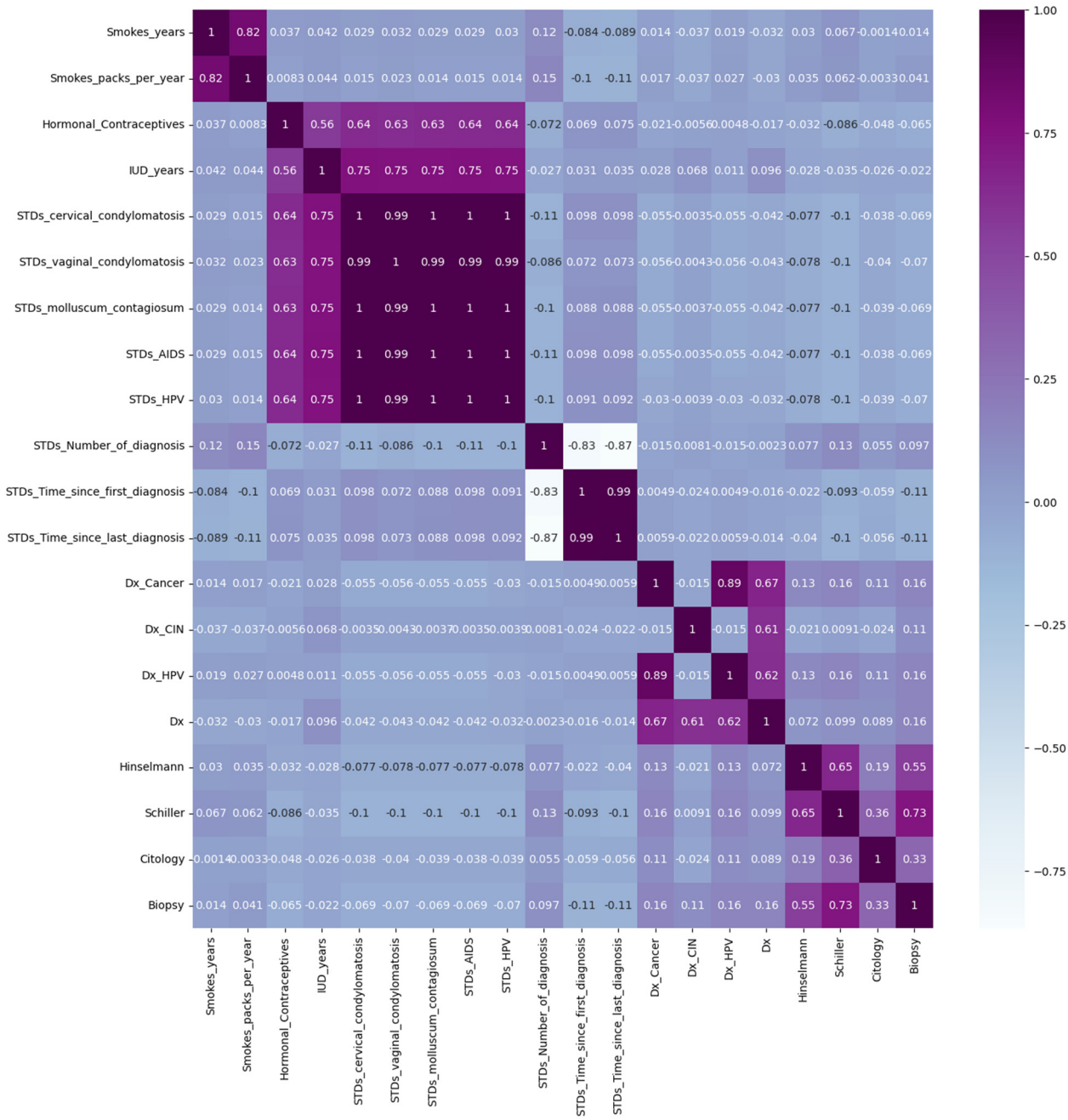

Fig. 4. Correlation of features (XGBoost + PCA).

3.3.4. Decision tree

Decision Tree is a well-known supervised machine learning technique for classification and regression tasks. It operates by recursively splitting the feature space into smaller parts and making judgments based on feature values. Each internal node represents a decision based on a feature value, and each leaf node represents the class label or regression value [33]. Equation (6) represents the DT classifier where A'' that gives the accurate decision.

$$A''(a_i) = [D_0(a_i) - D(a_i)] + K[\epsilon_0(a_i) - \epsilon(a_i)] + \sum_{j=1}^{b_i} x_{i+j} y_0(a_{i+j}) \tag{6}$$

Assuming a dataset $D = \{(a_1, b_1), (a_2, b_2), \ldots\ldots, (a_n, b_n)\}$, where a_i denotes the feature vector and b_i is the corresponding class label. The DT learns a function $f : x \rightarrow y$, where x is the feature space and y is the set of class labels [34].

3.3.5. Naive Bayes

Based on the Bayes theorem and supposing predictor independence, the Naive Bayes classifier is a probabilistic machine learning model. It is extensively employed in classification jobs, particularly in spam filtering and text classification [35]. The result of many complicated issues may be predicted simply and accurately by using NB [9]. It operates based on the Bayes theorem and takes a convinced attribute into account separately. Equation (7) represents the NB classifier.

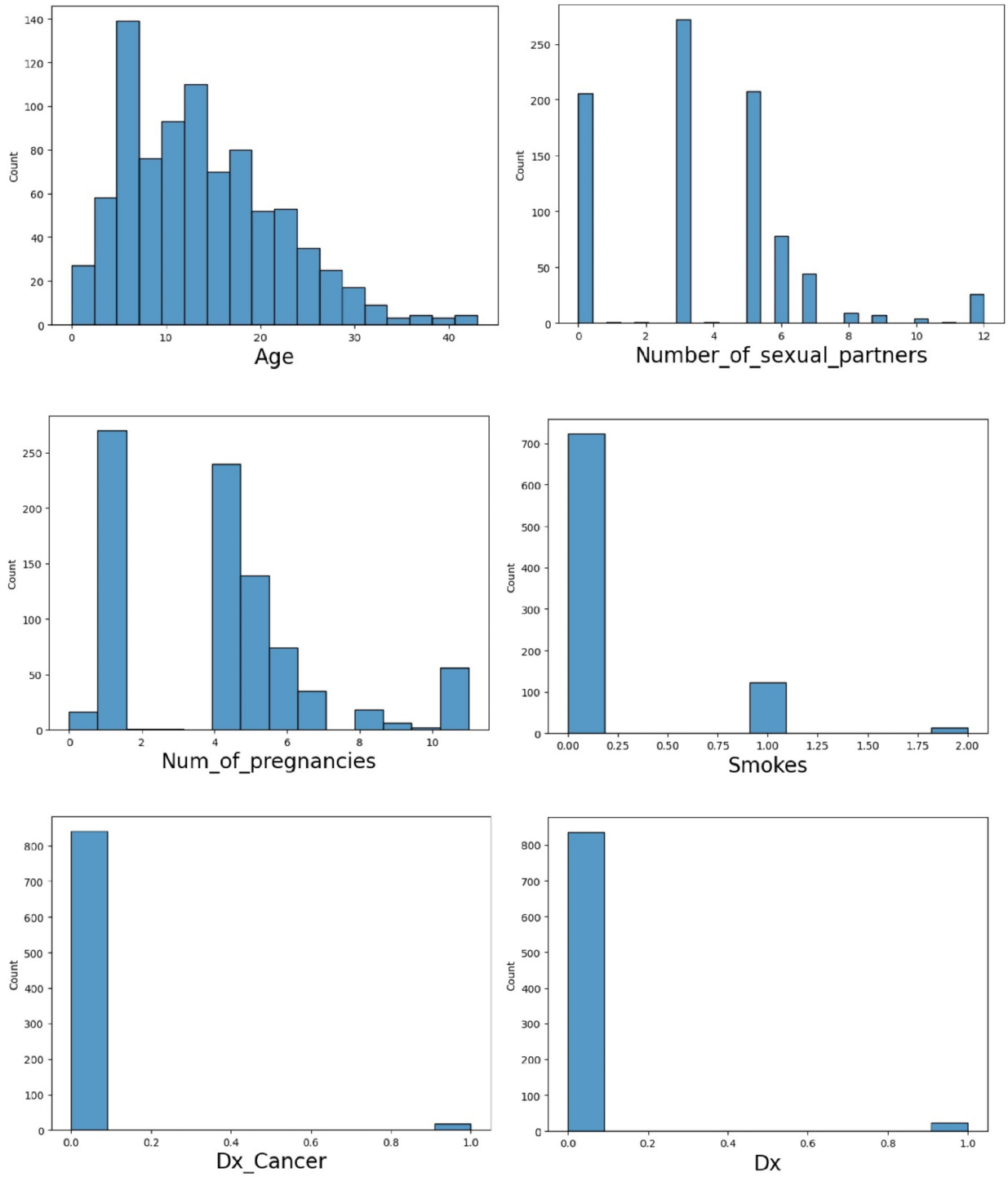

Fig. 5. Column Representation in Numerical.

$$x(A \mid B) = \frac{x(A, B)}{x(B)} = \frac{x(B \mid A)x(A)}{\sum_{y'=1}^{Z} x(B \mid A')x(B')} \tag{7}$$

Here,

- $x(A \mid B)$ is the probability of event A occurring given that event B has occurred.
- $x(B \mid A)$ is the probability of event B occurring given that event A has occurred.
- $x(A)$ and $x(B)$ are the probabilities of events A and B occurring, respectively.
- $x(A|B)$ denotes the chance of occurrence in the future and $x(A)$ is the prior probability.

3.3.6. Logistic regression

To establish a link between categorical dependent variables and independent variables, a typical supervised learning approach called LR computes the likelihood and predicts the result of a dependent variable with a category [36]. One common statistical technique is logistic regression. It represents the likelihood that an instance falls into a specific class. Logistic Regression relies on the

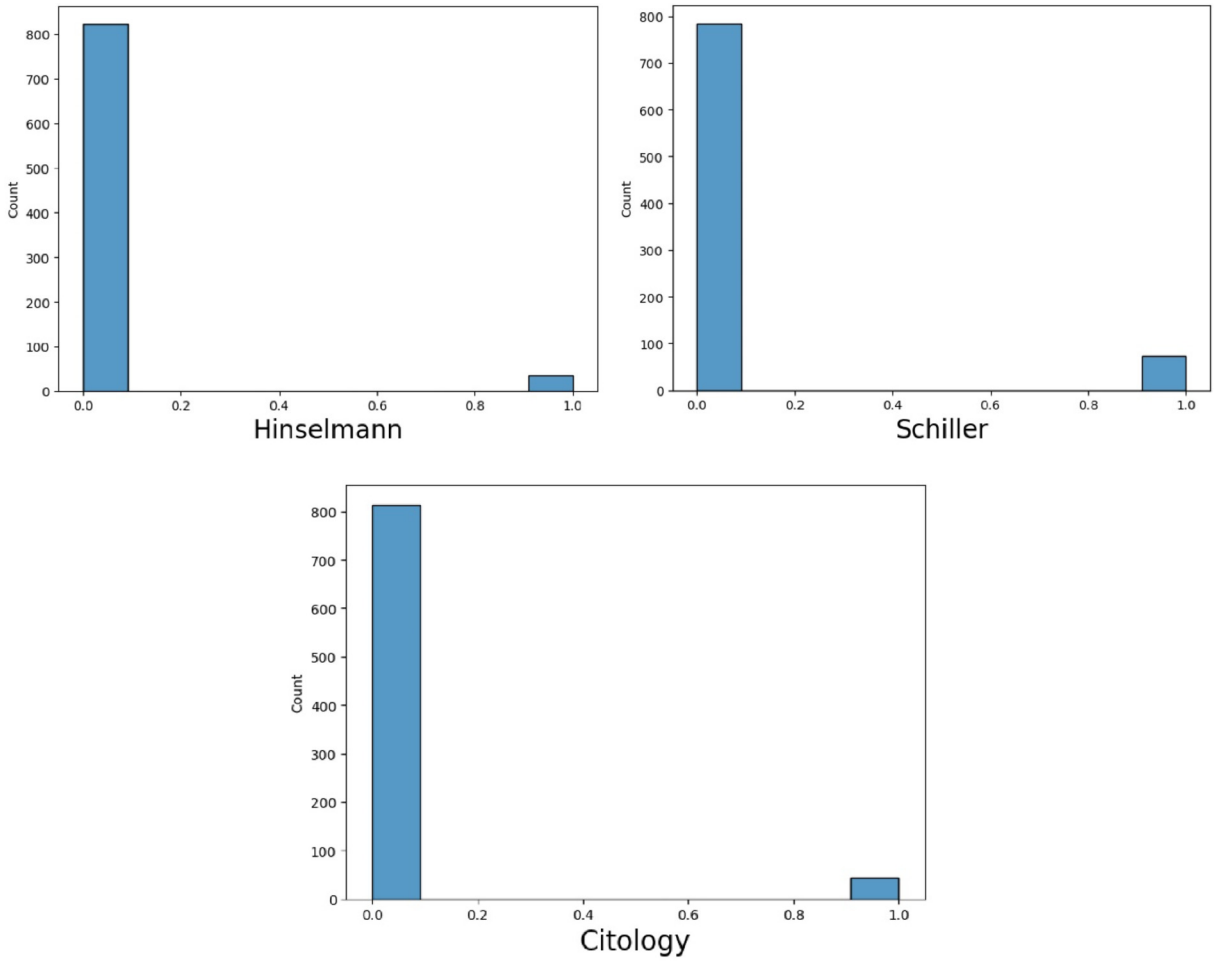

Fig. 5. (*continued*).

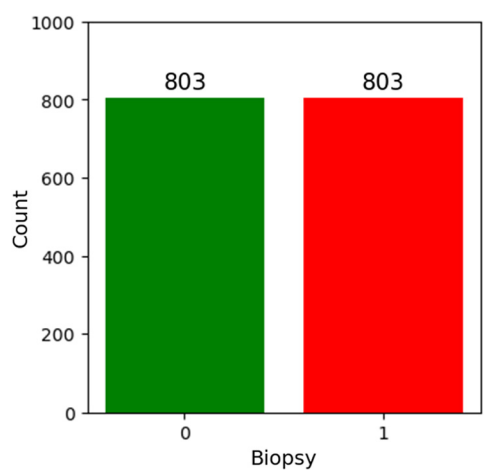

Fig. 6. Cancerous and non-cancerous patients after using Oversampling.

logistic function, which is often referred to as the sigmoid function [37]. Equation (8) represents the LR classifier.

$$P\left(a=\frac{1}{b}\right)=\frac{1}{1+e^{-W^{L}b+x}} \tag{8}$$

Here,

- $P(a=\frac{1}{b})$ is the probability of the target variable a being class 1 given input features.
- b represents the input feature vector.
- W is the weight vector.
- x is the bias or intercept term.
- e is the base of the natural logarithm.

3.3.7. Adaptive boost

Adaptive boosting is also known as AdaBoost. It's an enhancing method that is spread by the enhancing algorithm [38]. This algorithm aims to create a single strong classifier by combining several weak classifiers. The primary concept behind AdaBoost is to merge numerous weak classifiers into powerful classifiers. Each weak classifier focuses on the instances that the previous ones misclassified, gradually increasing overall classification performance [39]. Equation (9) represents the AB classifier.

$$B(z)=sign\left(\sum_{x=1}^{x}\theta_x b_x(z)\right) \tag{9}$$

Here,

- b_x denotes the xth vulnerable classifier.

3.3.8. Gradient boosting

A group of ML techniques called gradient boosting merges many insignificant learning models to produce a potent prediction model [38]. Gradient Boosting is an ensemble learning method that sequentially combines several weak learners to create a strong prediction model. The main principle of gradient boosting is to minimize the errors of the prior models to enhance the predic-

tive performance iteratively. Specifically, for classification tasks, the Gradient Boosting classifier is employed [40]. Equation (10) represents the GB classifiers.

$$G_x(a) = G_{x-1}(a) + \eta * h_x(a) \tag{10}$$

Here,

- $G_x(a)$ represents the prediction of the ensemble model at the xth iteration.
- $G_{x-1}(a)$ is the prediction of the ensemble model up to the $(x-1)^{th}$ iteration.
- η is the learning rate.
- $h_x(a)$ is the weak learner that is being fit to the residuals of the previous model.

3.3.9. Nearest centroid

The Nearest Centroid Classifier is a machine-learning classification algorithm that is simple and intuitive. It works on the principle of locating the centroid of each class in the feature space and then classifying new data points by assigning them to the class with the centroid closest to the data point in the feature space [41]. It entails predicting a new example's class label by identifying the class that the training dataset's centroid belongs to [38]. Equation (11) represents the NC classifiers.

$$\mu_i = \frac{1}{|A_i|} \sum_{x \epsilon A_x} y_x \tag{11}$$

Here,

- A_i is the number of samples in class μ_i.
- y_x represents the xth feature of the A_x^{th} sample in class μ_i.

3.3.10. Multilayer perceptron

Multilayer Perceptron is a type of feedforward artificial neural network that has multiple layers of nodes, including an input layer, one or more hidden layers, and an output layer. Every node in one layer is connected to every node in the next layer. The MLP can learn complicated correlations in data and is frequently used for classification and regression problems [42]. Its activation function is linear. Equation (12) represents the MLP classifier.

$$\hat{z} = f_2\left(\sum_{i=1}^{N_2} w_i^{(2)} * f_1\left(\sum_{j=1}^{N_1} w_{ij}^{(1)} * x_j + b_i^{(1)}\right) + b_2^{(2)}\right) \tag{12}$$

Here,

- $\hat{z}$ is the predicted output.
- f_1 and f_2 are the activation functions.
- N_1 and N_2 are the number of neurons.
- x_j represents the input features.
- $w_i^{(1)}$ and $w_i^{(2)}$ are the weights.
- $b_i^{(1)}$ and $b_2^{(2)}$ are the biases.

3.3.11. Voting classifier

A voting classifier is a type of ML classifier that trains a variety of models and selects the one with the greatest likelihood depending on the class of a category of interest. There are two categories: hard voting and easy voting [43]. In this proposed work both hard and soft voting is employed.
Equation (13) represents the hard voting.

$$hardvoting = argmax_j \sum_{i=1}^{N} 1\,(C_i predictsclassj) \tag{13}$$

Table 5
Environment setup of the system.

Resources	Details
CPU	Intel® Core™ i5-10400 CPU @ 2.90GHz
GPU	Intel® UHD Graphics 630
RAM	16 GB
Experimental tools	Google Colab, Jupyter Notebook

Equation (14) represents the soft voting.

$$softvoting = argmax_j \sum_{i=1}^{N} P\,(C_i predictsclassj) \tag{14}$$

Here,

- C_i represents individual classifiers.
- N represents the number of individual classifiers.
- 1 is the indicator function.
- P denotes the probability of C_i.

4. Results

A total of 20% and 80% of the data were used for testing and training the proposed system. Several ML classifiers, including SVM, RF, KNN, DT, NB, LR, AdB, GB, MLP, and NCC, are used to determine which model performs best. Use an ensemble machine learning classifier in addition to the best results. The voting classification method is used in this case. The best machine learning model among these ten classifiers is identified by using performance metrics such as f1-score, accuracy, recall, and precision.

4.1. System specification

Some resources were required to perform this research. The materials listed in Table 5 are used in the development of this suggested framework.

4.2. Confusion matrix

Accuracy shouldn't be the only factor that is taken into consideration when assessing a model. The confusion matrix is used to compute the metrics. A confusion matrix is made up of TP, TN, FP, and FN. It is a design for a table that enables the performance of the model to be visualized. True positive denotes a cancerous person who is correctly predicted to have cancer. TN stands for true negative which means a noncancerous person who was correctly predicted to be cancer-free. False positive, or FP, refers to the mistaken diagnosis of cancer in a noncancerous individual. False negative, or FN, is a result that suggests a malignant person is cancer-free. FN should be maintained as minimal as possible because it is obviously the most critical element. Those performance metrics were used in this study to assess the performance of the model we've proposed. Following is a definition of those metrics utilizing their basic representations:

i. Accuracy: The proportion of the model's accurate predictions that were made out of all the predictions and it is defined by Equation (15).

$$\frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

ii. Sensitivity: The model's capacity to accurately detect cervical cancer in individuals. A sensitivity of 100% means that all cervical cancer cases were correctly predicted by selected model and it is defined by Equation (16).

$$\frac{TP}{TP+FN} \tag{16}$$

iii. Specificity: This measure assesses how well the model can identify individuals who will never get cervical cancer and it is defined by Equation (17).

$$\frac{TN}{TN+FP} \tag{17}$$

iv. Precision: This metrics count's the percentage of cases of cervical cancer that the model correctly identified. Precision measures how many cervical cancer cases the model correctly predicted and it is defined by Equation (18).

$$\frac{TP}{TP+FP} \tag{18}$$

v. F-Measure: The model's sensitivity and precision are combined to form the F-Measure which is known as the harmonic mean of the two and it is defined by Equation (19).

$$\frac{TP}{TP+\frac{1}{2}(FP+FN)} \tag{19}$$

Fig. 7 shows the confusion matrix for all ML classifiers, here the best features are selected byPCA and XGBoost feature selection method.

The evaluation metrics sensitivity, specificity, precision, and F-Measure are particularly essential in the context of binary classification problems. They provide essential details about a classifier's performance and are essential in figuring out how well the model functions overall.

These metrics provide a comprehensive view of the classifier's performance in various aspects:

- Sensitivity and specificity focus on the correct identification of positive and negative instances.
- Precision focuses on the accuracy of positive predictions.
- F-Measure provides a balanced assessment that takes into account both false positives and false negatives.

Table 6 shows the model evaluation of different machine-learning classifiers. Where the best combination of XGBoost + PCA+ Random Oversampling + Random Forest achieved 100% sensitivity with 97.93%, 97.88%, and 98.93% specificity, precision, and F-measure respectively and the worst combination of SelectKbest + PCA+ Random Oversampling + K-Nearest Neighbor achieved 81.33% sensitivity with 96.45%, 97.14%, and 96.38% specificity, precision, and F-measure respectively.

Accuracy is an important evaluation metric in machine learning because it provides a simple measure of how well a model performs overall. It denotes the proportion of correctly predicted instances to the total number of instances in a dataset. Table 7 shows the accuracy of ten machine-learning classifiers with all combinations of feature selection models and Random Oversampling. With XGBoost + PCA, Logistic Regression achieved the best at 97.67% accuracy and Naïve Bayes achieved the worst at 91.47% accuracy. On the other hand, with XGBoost + PCA + Random Oversampling, Random Forest achieved the best 98.94% accuracy and Naïve Bayes achieved the worst 91.41% accuracy.WithSelectKBest + PCA, Logistic Regression and Nearest Centroid Classifier achieved the best 96.90% accuracy and Naïve Bayes achieved the worst 86.43% accuracy. On the other hand, with SelectKBest + PCA + Random Oversampling, Random Forest achieved the best 96.45% accuracy and Nearest Centroid Classifier achieved the worst 90.54% accuracy.WithXGBoost + SelectKBest + PCA, Logistic Regression achieved the best 97.29% accuracy and Naïve Bayes achieved the worst 92.25% accuracy. On the other hand, with XGBoost + SelectKBest + PCA + Random Oversampling, Random Forest achieved best 98.69% accuracy and Nearest Centroid Classifier achieved worst 89.60% accuracy.

Table 8 shows the accuracy of ensemble machine-learning classifiers on Gradient Boosting, Random Forest, and Multilayer Perceptron with all combination of feature selection model and Random Oversampling. Here the combination of XGBoost + PCA + Random Oversampling with hard voting on Random Forest and Multilayer Perceptron achieved best 99.19% accuracy.

Fig. 8 illustrates the accuracy of hard voting and soft voting with the combination of XGBoost + PCA + Random Oversampling where the hard voting on Random Forest and Multilayer Perceptron achieved the best 99.19% accuracy and the soft voting on Gradient Boosting and Random Forest achieved the worst 98.63% accuracy.

4.3. ROC & AUC

ROC curve is a method for displaying a model's efficiency. It is a thorough index that reflects the sensitivity and specificity of continuous variables. The curve depicts the connection between sensitivity and specificity. The area under the ROC curve (AUC) is frequently used to assess the performance of models, also known as the ROC curve. The overall two-dimensional surface area of the ROC curve is measured by AUC. The AUC value increases as the model's performance improves.

Fig. 9 illustrates the ROC curve and AUC value of all machine-learning models with the combination of XGBoost + PCA + Random Oversampling where the best AUC value of 96.14% was achieved by the Random forest and the lowest AUC value of 85.33% was achieved by the Adaptive Boost Classifier.

Fig. 10 illustrates the ROC curve and AUC value of all machine-learning models with the combination of SelectKBest + PCA + Random Oversampling where the best AUC value of 92.11% was achieved by the Support Vector Machine and the lowest AUC value of 72.92% achieved by the K-Nearest Neighbor Classifier.

Fig. 11 illustrates the ROC curve and AUC value of all machine-learning models with the combination of XGBoost + SelectKBest + PCA + Random Oversampling where the best AUC value of 94.55% achieved by the Naïve Bayes classifiers and the lowest AUC value of 78.28% achieved by the Decision Tree Classifier.

5. Discussion

In this work, we evaluated our proposed model's efficacy in predicting the risk of cervical cancer with the most recent approaches published by Tanimu J. et al. [9], Abdoh et al. [44], and Ijaz et al. [45]. Our model showed encouraging results in terms of accuracy, sensitivity, and specificity by combining sampling techniques with machine learning algorithms. Abdoh et al. [44] used SMOTE-RF, SMOTE-RF-RFE, and SMOTE-RF-PCA techniques, whereas Ijaz et al. [45] used DBSCAN, SMOTETomek, RF, iForest, and SMOTE in various combinations. Additionally, Tanimu J. et al. [9] focused on RFE, LASSO, SMOTETomek, and DT algorithms.

However, Table 9 shows our approach performed better than these methods in terms of accuracy, achieving an unusual 99.19% when using a voting technique on Random Forest (RF) and Multilayer Perceptron (MLP). Furthermore, our model achieved competitive accuracies of 99.00% and 98.63% in various combinations, including ROS+XGBoost+PCA+RF and ROS+XGBoost+PCA+MLP. At this time, we achieved a significant accuracy of 98.57% using ROS+XGBoost+PCA+GB. Furthermore, our model shows higher sensitivity than the previously mentioned methodologies.

Table 6
Model Evaluation of different ML classifiers using the combination of different feature selection approaches.

Data	Model	Sensitivity	Specificity	Precision	F-Measure
XGBoost + ROS + PCA	SVM	92.52%	95.37%	95.52%	94.00%
	RF	100%	97.93%	97.88%	98.93%
	KNN	100%	97.22%	97.14%	98.55%
	DT	100%	96.98%	96.89%	98.42%
	NB	93.81%	89.26%	88.67%	91.71%
	LR	93.96%	93.09%	93.03%	93.49%
	AdB	99.74%	97.21%	97.14%	98.42%
	GB	100%	97.22%	97.14%	98.55%
	MLP	100%	97.10%	97.01%	98.48%
	NCC	87.06%	95.03%	95.52%	91.09%
KBest + ROS + PCA	SVM	91.79%	95.82%	96.01%	93.85%
	RF	98.44%	94.62%	94.40%	96.38%
	KNN	81.33%	96.45%	97.14%	88.54%
	DT	97.78%	93.46%	93.15%	95.41%
	NB	93.82%	87.82%	86.92%	90.24%
	LR	91.77%	95.57%	95.77%	93.72%
	AdB	97.29%	94.10%	93.90%	95.56%
	GB	97.41%	93.88%	93.65%	95.49%
	MLP	96.13%	95.90%	95.89%	96.01%
	NCC	86.78%	95.15%	95.64%	91.00%
XGBoost + KBest + ROS + PCA	SVM	93.35%	94.33%	94.40%	93.87%
	RF	100%	97.45%	97.38%	98.68%
	KNN	100%	97.33%	97.26%	98.61%
	DT	100%	95.14%	94.89%	97.38%
	NB	93.36%	89.74%	89.29%	91.28%
	LR	94.24%	93.69%	93.65%	93.94%
	AdB	100%	95.82%	95.64%	97.77%
	GB	100%	97.10%	97.01%	98.48%
	MLP	100%	97.22%	97.14%	98.55%
	NCC	85.41%	94.92%	95.52%	90.18%

Table 7
The Accuracy of all ML classifiers.

Accuracy Model Name	*XGBoost + PCA*	*XGBoost + PCA + ROS*	*KBest + PCA*	*KBest + PCA + ROS*	*KBest + XGBoost + PCA*	*KBest + XGBoost + PCA + ROS*
SVM	96.51%	93.90%	93.80%	93.71%	95.35%	93.84%
RF	96.12%	98.94%	92.64%	96.45%	96.12%	98.69%
KNN	96.51%	98.57%	92.64%	87.42%	96.90%	98.63%
DT	94.19%	98.44%	92.64%	95.52%	93.41%	97.45%
NB	91.47%	91.41%	86.43%	90.60%	92.25%	91.47%
LR	97.67%	93.52%	96.90%	93.59%	97.29%	93.96%
AdB	94.57%	98.44%	92.64%	95.64%	94.96%	97.82%
GB	94.96%	98.57%	93.02%	95.58%	96.12%	98.51%
MLP	97.29%	93.51%	92.64%	96.01%	96.90%	98.57%
NCC	96.90%	90.66%	96.90%	90.54%	96.12%	89.60%

Table 8
The Accuracy of Ensemble (Hard Voting& Soft Voting) ML classifiers.

Accuracy			
Model Name	*XGBoost +PCA +ROS + Voting (Hard)*	*KBest +PCA +ROS + Voting (Hard)*	*KBest +XGBoost +PCA +ROS + Voting (Hard)*
GB + RF	99.07%	95.89%	99.00%
GB + MLP	99.00%	95.89%	99.00%
RF + MLP	99.19%	96.26%	99.00%
GB + RF + MLP	98.82%	95.89%	98.75%
Accuracy			
Model Name	*XGBoost + PCA + ROS + Voting (Soft)*	*KBest + PCA + ROS + Voting (Soft)*	*KBest + XGBoost + PCA + ROS + Voting (Soft)*
GB + RF	98.63%	95.95%	98.63%
GB + MLP	98.75%	95.58%	98.63%
RF + MLP	99.00%	96.51%	98.69%
GB + RF + MLP	98.88%	95.89%	98.63%

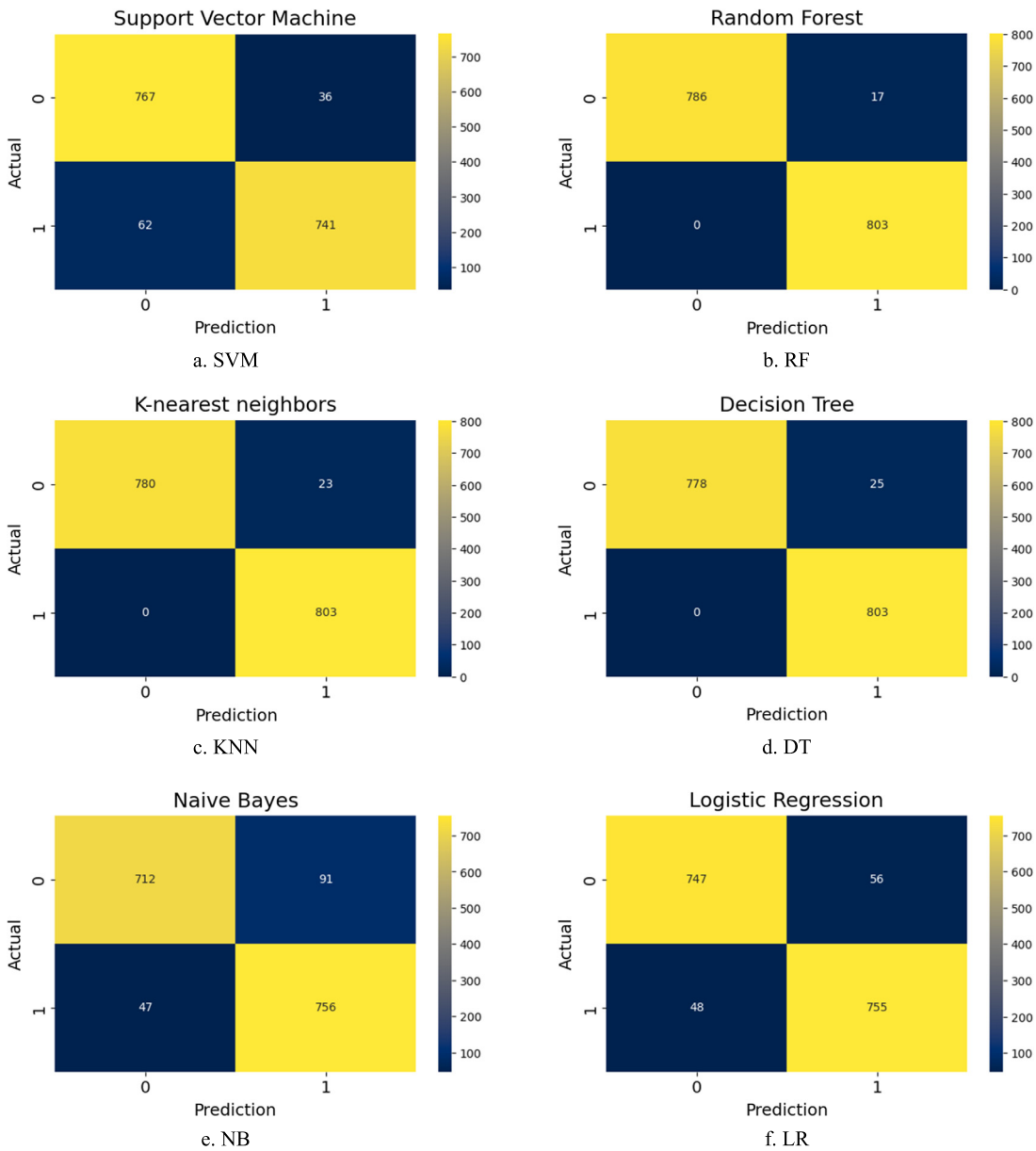

Fig. 7. Confusion Metrics of (a) SVM (b) RF (c) KNN (d) DT (e) NB (f) LR (g) NCC (h) AdB (i) GB (j) MLP.

Table 9
Comparison results of our model with past studies.

Studies	Method	Features count	Accuracy	Sensitivity	Specificity
Abdoh et al. [44]	SMOTE-RE	30	96.06%	94.55%	97.51%
	SMOTE-RE-RFE	18	95.87%	94.42%	97.26%
	SMOTE-RE-PCA	11	95.74	94.16%%	97.76%
Ijaz et al. [45]	DBSCAN+SMOTETomek+RF	10	97.72%	97.43%%	98.01%
	DBSCAN+SMOTE+RF	10	97.22%	96.43%	98.01%
	iForest+SMOTETomek+RF	10	97.50%	97.91%	97.08%
	iForest+SOMET+RF	10	97.58%	97.45%	97.58%
Tanimu, J.J [9]	RFE+DT	20	97.65%	85.71%	98.72%
	LASSO+DT	10	96.47%	71.43%	98.71%
	RFE+SMOTETomek+DT	20	98.82%	100%	98.71%
	LASSO+ SMOTETomek+DT	10	92.94%	85.71%	93.59%
Our Proposed Model	ROS+XGBoost+PCA+RF	20	98.94%	100%	97.93%
	ROS+XGBoost+PCA+GB	20	98.57%	100%	97.22%
	ROS+XGBoost+PCA+MLP	20	98.51%	100%	97.10%
	ROS+KBest+PCA+RF	20	96.45%	98.44%	94.62%
	ROS+KBest+PCA+AB	20	95.64%	97.29%	94.10%
	ROS+KBest+PCA+MLP	20	96.01%	96.13%	95.90%
	ROS+XGBoost+PCA+ Hard Voting(RF+MLP)	**20**	**99.19%**	**100%**	**–**

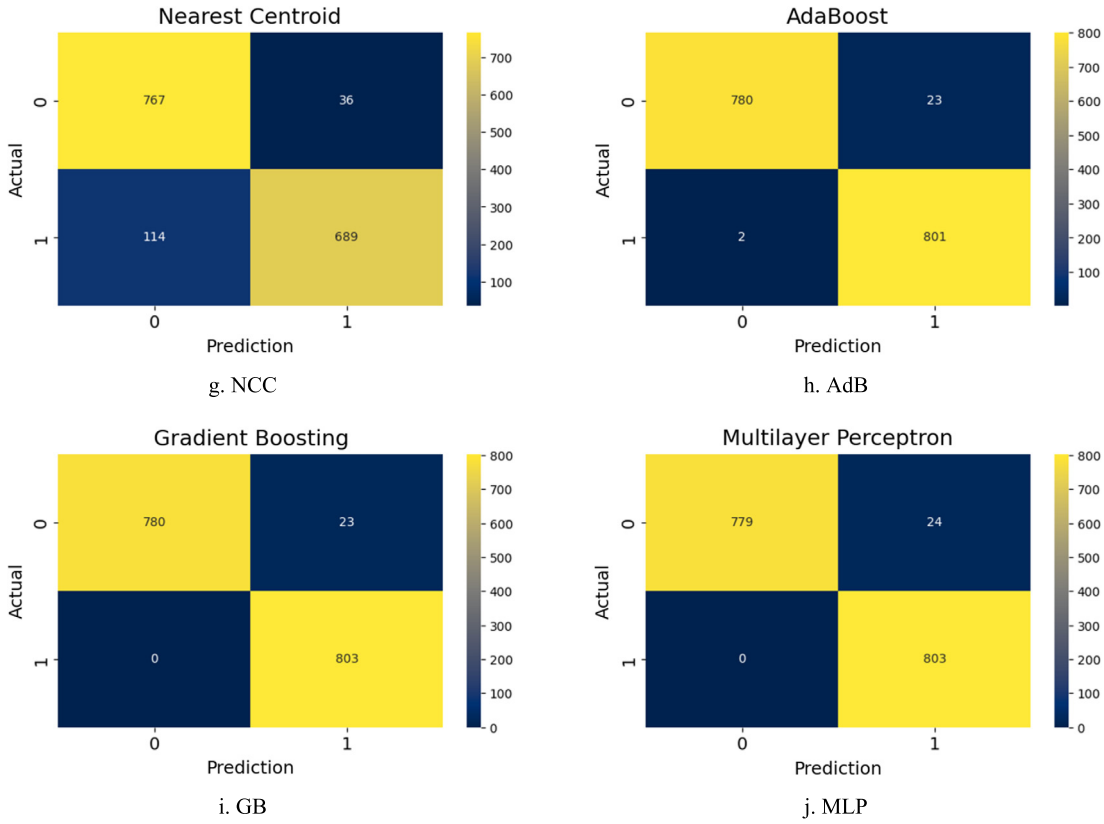

Fig. 7. (*continued*).

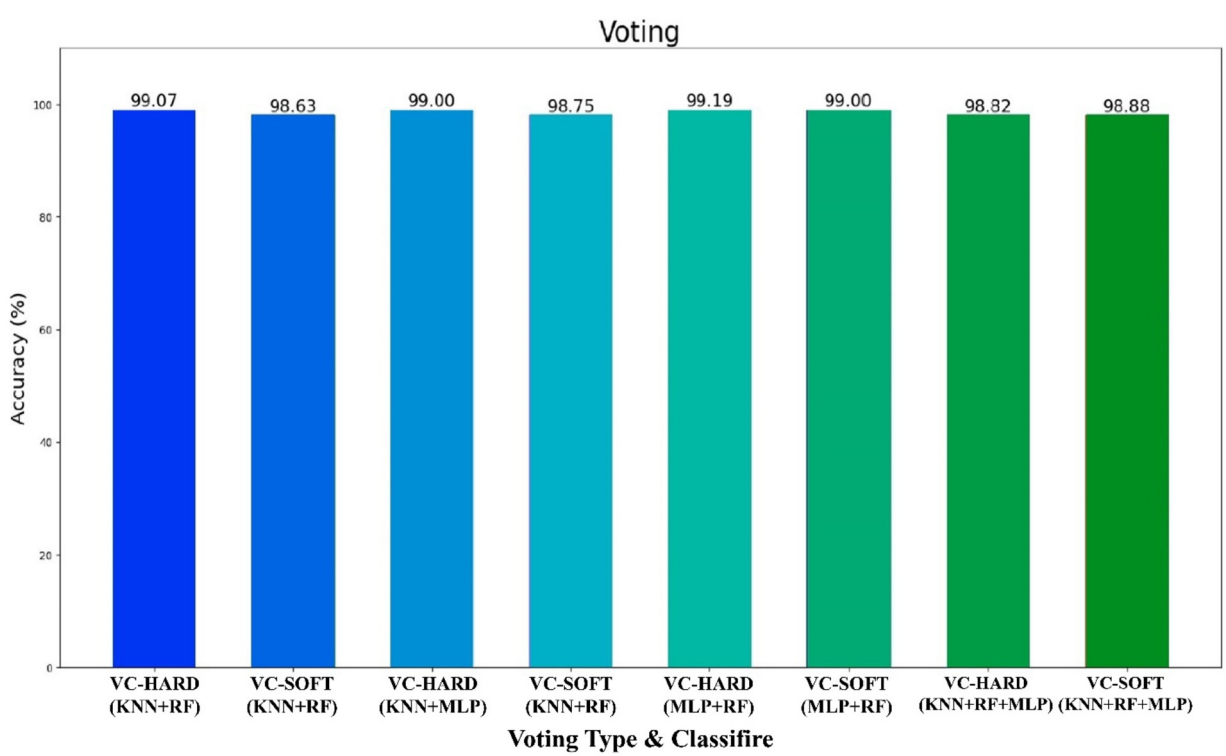

Fig. 8. Accuracy of Voting Classifiers.

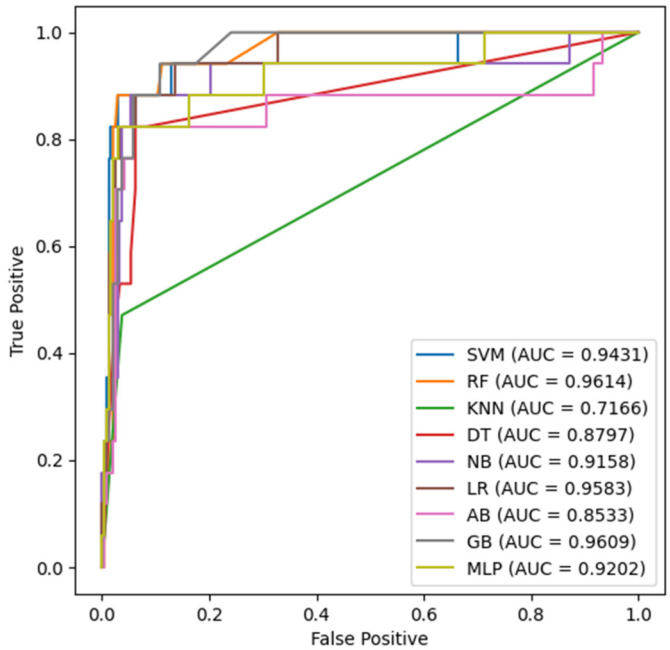

Fig. 9. ROC curve and AUC (XGBoost+ROS+PCA).

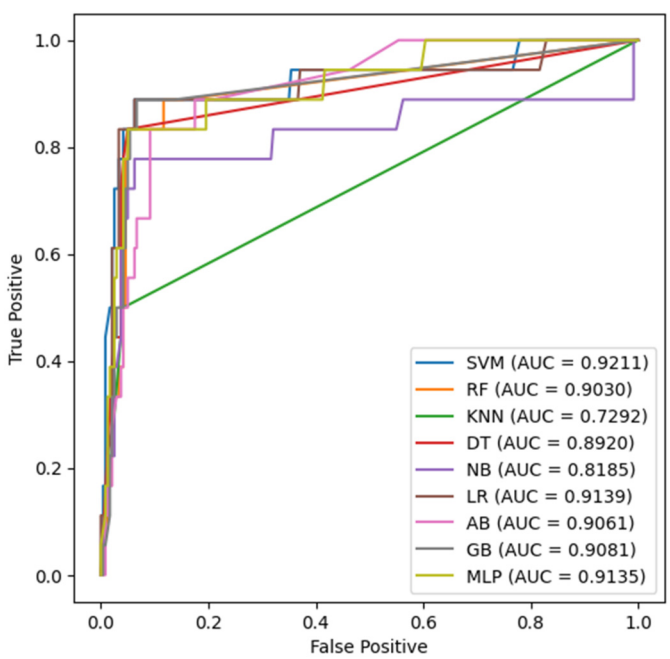

Fig. 10. ROC curve and AUC (KBest+ROS+PCA).

6. Conclusion

Cervical cancer is now recognized as a major cause of death among women. The World Health Organization (WHO) reports that more than 85% of all cervical cancer cases occur in developing countries. Thanks to machine learning, we can identify factors that increase the risk of this disease in women. In our study, we tested ten different machine learning models and used an Ensemble ML classifier (hard voting) to predict cervical cancer based on patient data regarding risk factors. By balancing the classes in our dataset, we significantly improved the accuracy of our predictions. We found that random oversampling particularly enhanced our classification results. Compared to previous methods, our recommended model showed great improvement in experimental tests.

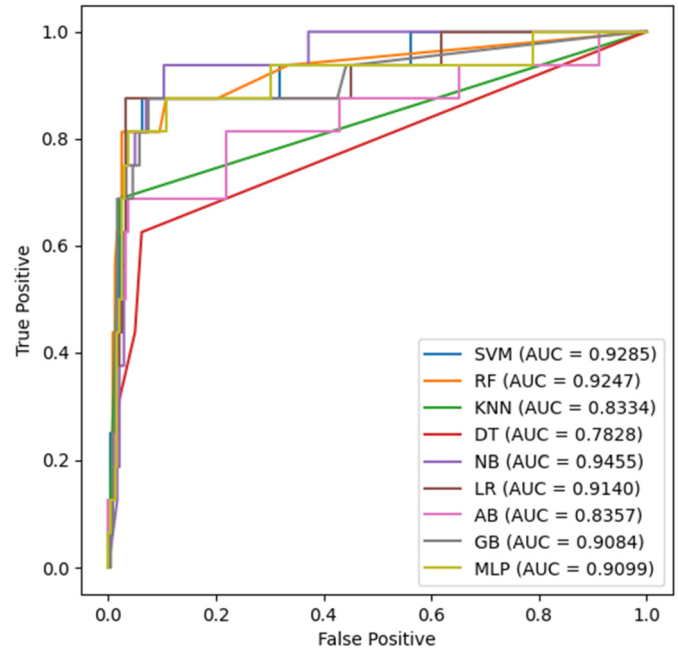

Fig. 11. ROC curve and AUC (XGBoost+KBest+ROS+PCA).

Our best-performing model, utilizing hard voting with a combination of MLP, RF, XGBoost, and PCA, achieved an accuracy of 99.19% and 100% sensitivity. Although we also explored feature selection techniques like SelectKbest and a combination of XGBoost and SelectKbest, they didn't perform as well as XGBoost alone with PCA. In future research, we will gather a larger dataset to improve the performance of our system. We'll also look into more advanced class balancing algorithms and ensemble-based classifiers like boosting and bagging to create an online screening tool.

Human and animal rights

The authors declare that the work described has not involved experimentation on humans or animals.

Informed consent and patient details

The authors declare that this report does not contain any personal information that could lead to the identification of the patient(s) and/or volunteers.

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Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship.

Declaration of competing interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

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